Blackmon Lab

Evolutionary Biology

Relationship between Sexual Dimorphism and Mating System in Galliformes

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Background

Mating systems in Galliformes range from strict monogamy to extreme polygamy, and this variation is often discussed in the context of sexual selection and the evolution of conspicuous traits (e.g., elaborate plumage and display behaviors). At the same time, Galliformes show broad differences in sexual dimorphism, which we estimate using PC2 from eight morphological traits (Figure 1). Prior work on Galliformes frequently emphasizes dramatic, highly ornamented polygamous taxa (e.g., peafowl and grouse), but fewer studies explicitly test, within a phylogenetic framework, whether different mating systems are associated with different evolutionary “targets” for sexual dimorphism across the clade. **We predicted that the mating system is associated with predictable evolutionary shifts in sexual dimorphism, with strict monogamy favoring lower dimorphism and extreme polygamy favoring higher dimorphism.** We combine explicit modeling of mating system evolution on the tree (Figure 2) with model-based tests of whether sexual dimorphism evolves toward different regime-specific optima (OUM; Figure 3), rather than relying only on visual patterns or non-phylogenetic comparisons.

Study Design & AI

We compiled morphological and behavioral data for 50 Galliformes species, including body shape measurements captured by principal component analysis and mating system classifications. Mating systems were coded on a five-level scale from strict monogamy (0) to extreme polygamy (4), representing increasing intensity of sexual selection. We focused on PC2 as our morphological trait because it captures body shape variation independent of overall size (PC1), allowing us to test whether sexual selection drives shape evolution specifically rather than simply favoring larger individuals. PC2 values in our dataset ranged from -1.85 in species like *Penelope obscura* to +3.03 in *Centrocercus urophasianus*, indicating substantial morphological diversity across the clade. We paired these data with a time-calibrated phylogenetic tree pruned to match species with complete morphological and behavioral information, enabling phylogenetic comparative methods that account for shared evolutionary history among related species.

Model Fitting

We employed two phylogenetic comparative approaches. First, we fit a Markov k-state (MK) model to the discrete mating system trait to summarize how mating systems change across the tree, and we visualized one possible history as a stochastic map (Figure 2). Second, we fit continuous-trait models to PC2 to test different evolutionary processes. Brownian Motion (BM) represents evolution as a random drift, while Ornstein-Uhlenbeck (OU) represents evolution with attraction towards an optimum. Finally, we fit a multi-optima OU model (OUM) that estimates a separate optimum. Finally, the multi-optima OU (OUM) model treated each mating system category as a distinct selective regime, estimating separate optimal (θ) for sexual dimorphism in each mating system category. We compared model support using AIC. Lower values indicate better support.

Data

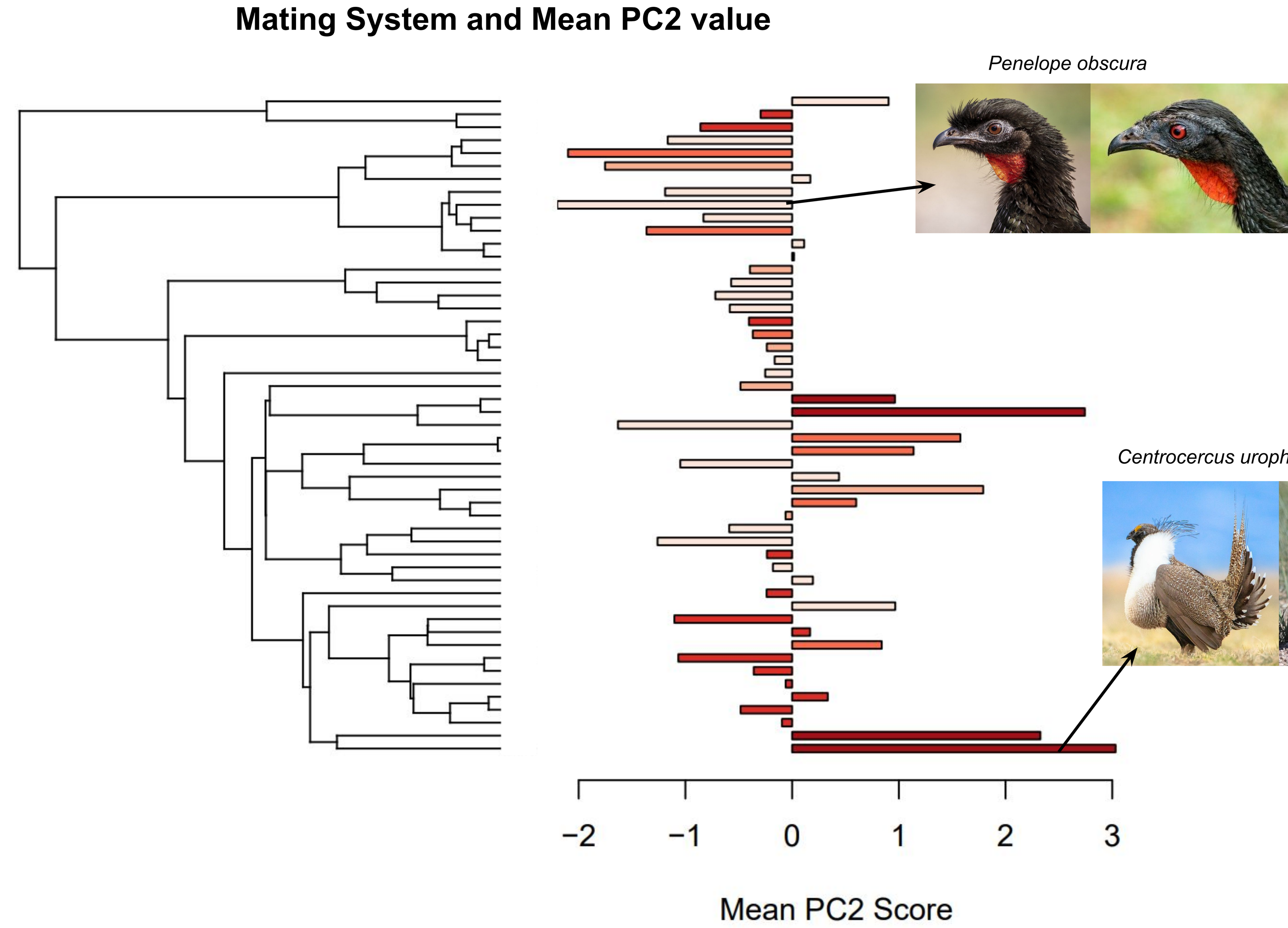

Figure 1. PC2 (sexual dimorphism). This plot shows PC2 scores for species in our dataset. Large positive values indicate high dimorphism while high large negative values indicate low levels of dimprphism.

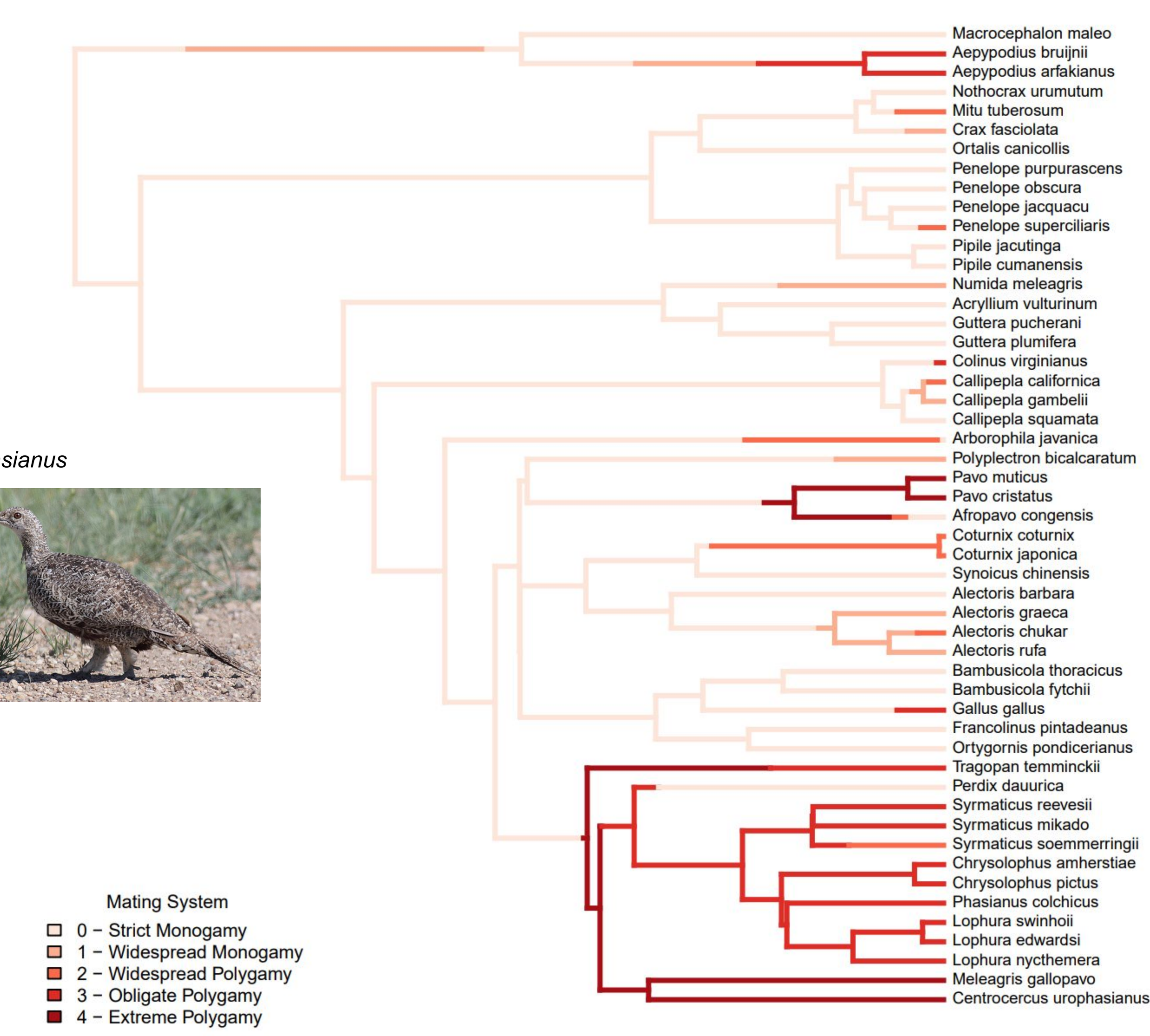

Figure 2. Stochastic Map. This plot shows one possible history for the evolution of mating systems across Galiformes.

Results

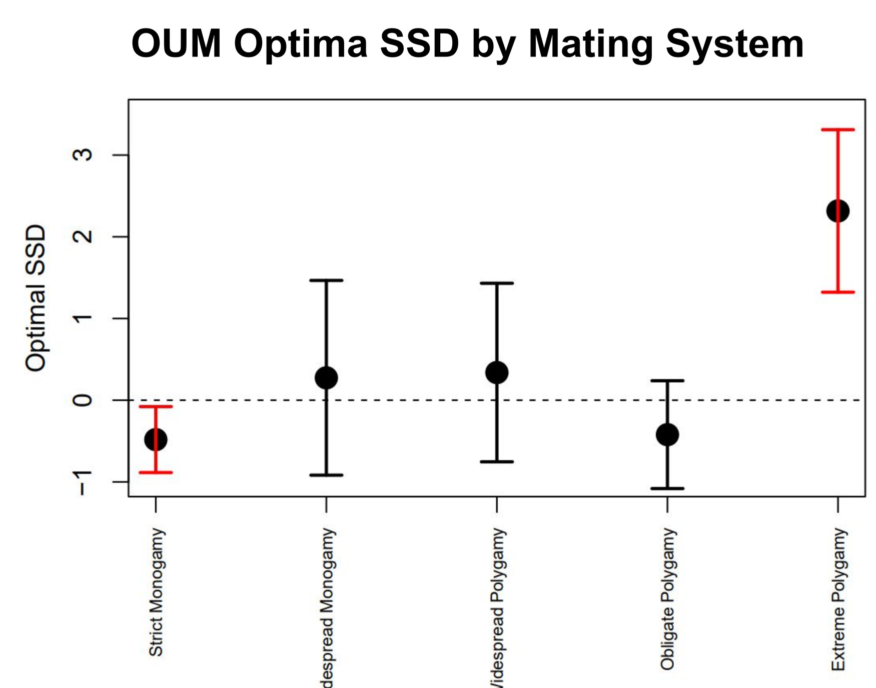

Figure 3. OUM optima by Mating System. Strict Monogamy excludes zero on the negative side and extreme polygamy excludes zero on the positive side, indicating significantly different optima SSD values at the mating-system.

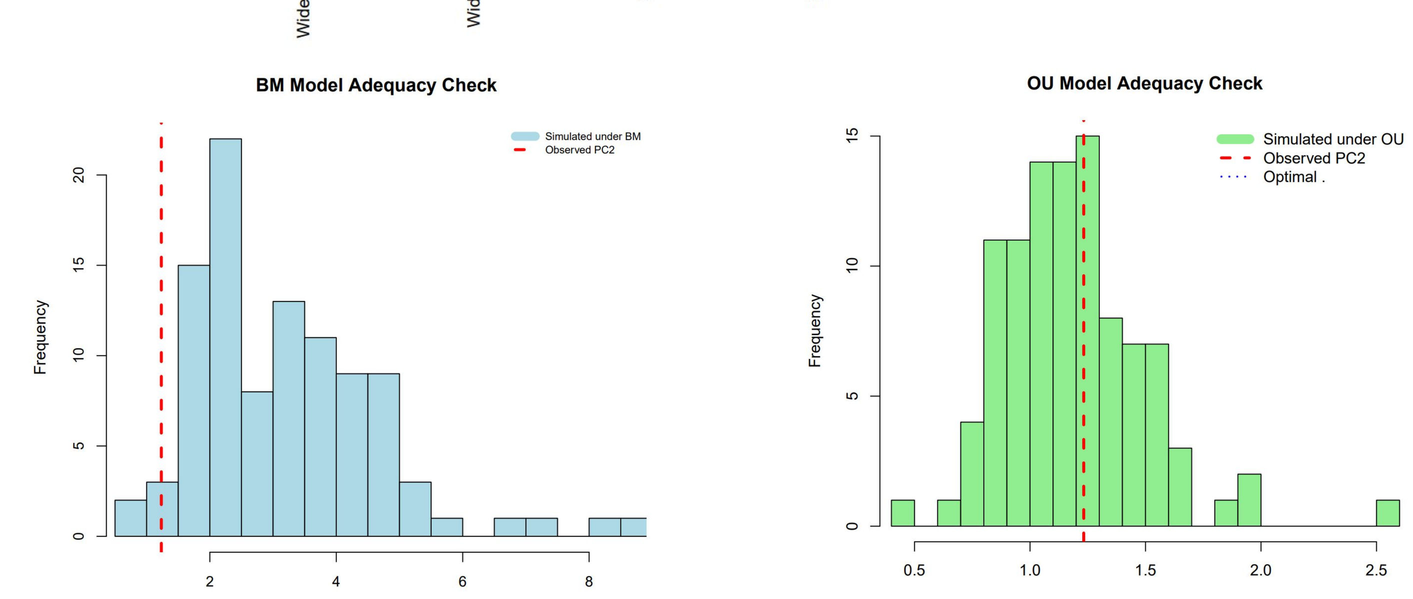

Figure 4. Model adequacy checks for PC2 under BM and OU Models. Histogram show simulated trait variances under each fitted model, with the observed variance marked by a dashed line. AIC for BM = 158.864404; AIC for OU = 148.324677.

Discussion

Figure 4 explains why we moved beyond BM and toward OU-type models. The BM model had a higher AIC than the OU model, meaning the OU provides a better explanation of PC2 evolution once model complexity is considered. Biologically, this supports the idea that sexual dimorphism does not evolve by unconstrained random drift alone; instead, it shows evidence of being pulled toward an optimum. Under the OUM framework in Figure 3, we estimated an optimal sexual dimorphism value (θ) for each mating system regime and plotted confidence intervals around those optima. Strict Monogamy has a negative optimum that excludes zero, consistent with selection favoring reduced dimorphism, whereas Extreme Polygamy has a positive optimum that excludes zero, consistent with selection favoring increased dimorphism. This is the strongest evidence in the dataset that mating system extremes are associated with different evolutionary “targets” for sexual dimorphism.

Conclusions

- Galliformes show wide variation in sexual dimorphism (PC2) and mating system.
- OU fits PC2 better than BM (AIC_OU = 148.324677 < AIC_BM = 158.864404), supporting constrained evolution.
- OUM results indicate regime-specific optima: strict monogamy favors lower (negative) dimorphism, and extreme polygamy favors higher (positive) dimorphism (Figure 3).